National University of Singapore

Electrical and Computer Engineering

EE5311: Differentiable and Probabilistic Computing

Neural ODE Modelling of Drone Ground Effect During Vertical Landing

A Grey-Box Study in Differentiable Physics and SciML

CA1 Project Blog

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1 Abstract

This report presents a grey-box modelling study for vertical drone landing under strong near-ground aerodynamic distortion. The target phenomenon is the "ground effect," whereby the rotor downwash interacts with the ground plane, modifying lift and damping near touchdown. In practice, this effect renders conventional landing models unreliable during the safety-critical final descent.

A data-driven Neural ODE framework was investigated to model the unknown aerodynamic residuals while retaining the known mechanics of gravity, thrust, and drag. Three models were compared: a fixed-physics white-box baseline, a pure black-box neural dynamics model, and a grey-box universal differential equation (UDE) with a structured near-ground force term.

The resulting grey-box UDE achieved state-of-the-art accuracy on held-out test data, yielding a Global Altitude RMSE of ∼0.018 m—a significant improvement over the white-box baseline (∼0.068 m) and the pure black-box model (∼0.062 m). In the critical near-ground regime ($h < 0.5$ m), the UDE reduced the error by approximately 77% relative to the baseline. Furthermore, analysis of training dynamics reveals that the UDE framework exhibits superior convergence stability, effectively mitigating the heavy stochastic oscillations observed in the black-box counterpart. Diagnostic visualizations of energy decay and phase portraits further confirm that the structured Neural ODE provides a robust compromise between high-precision trajectory reconstruction and physical plausibility.

2 Introduction

Accurate autonomous landing remains one of the most delicate phases of rotorcraft operation [4]. Whereas high-altitude vertical motion is often captured adequately by thrust, drag, and gravity alone, the final metre before touchdown is materially affected by aerodynamic interaction with the ground plane [1]. This interaction modifies the local pressure field, changes the effective downwash, and introduces additional lift-like behaviour that is difficult to capture with low-order parametric models.

Traditional white-box landing models fail because the final approach is governed by a partially known mechanism. If the ground effect is omitted, the model typically predicts an overly aggressive descent or an early impact. If empirical correction terms are introduced without sufficient structure, the resulting model may become numerically fragile or physically opaque. This tension makes the problem well suited to differentiable modelling [3].

Neural Ordinary Differential Equations (Neural ODEs) were selected as the primary modelling paradigm because they permit a continuous-time residual term to be embedded directly inside an ODE solver [2]. In this study, we propose a **structured Grey-box UDE** that preserves known vertical mechanics while using a neural network to augment only the unknown ground-effect residuals. Unlike unconstrained black-box models, our formulation employs physically-inspired gating and clamping mechanisms to ensure numerical stability and interpretability. This approach is particularly appropriate for UAV landing because the state evolves continuously and the latent force field is strongly state-dependent, localized, and inherently non-linear [2, 3].

3 Problem Formulation

3.1 State Definition and Sign Convention

The system state is defined by the drone's vertical position and motion during the landing phase. In the provided dataset, the altitude $h(t)$ is measured upward from the ground plane (where $h = 0$ at touchdown). Crucially, the vertical velocity $v(t)$ is defined as **positive downward**. With this sign convention, the kinematic relationship between altitude and velocity is expressed as:

$$\dot{h}(t) = -v(t) \tag{1}$$

This formulation ensures that as the drone descends towards the ground ($v > 0$), the rate of change of altitude is negative ($\dot{h} < 0$), accurately reflecting the physical landing process.

3.2 Vertical Force Balance

The vertical dynamics of the rotorcraft are governed by Newton's Second Law, incorporating gravitational, propulsive, and aerodynamic forces. The total vertical force balance is formulated as the following second-order dynamic equation:

$$m\dot{v}(t) = mg - (T(t) + c_{drag}v(t) + F_{\text{ge}}(h, v)) \tag{2}$$

In our experimental setup, the following physical constants are fixed to match the parameters of the trajectory generator:

- $m = 1.5\,\text{kg}$: The total vehicle mass.
- $g = 9.81\,\text{m/s}^2$: The acceleration due to gravity.
- $c_{drag} = 0.8$: The linear drag coefficient representing air resistance during descent.
- $T(t)$: The time-varying thrust command exported from the flight controller for each trajectory.

The term $F_{\text{ge}}(h, v)$ represents the **Ground Effect**, an unknown state-dependent aerodynamic force that arises from the interaction between the rotor downwash and the ground surface. Identifying and modeling this residual force is the primary objective of the Neural ODE training.

3.3 Properties of the Unknown Near-Ground Term

Based on rotorcraft aerodynamic principles, the ground-effect force F_{ge} is non-linear and highly localized. In this study, we assume it satisfies three qualitative properties:

1. **Spatial Localization**: The force is negligible far from the ground. Our implementation defines a critical threshold at $h_c = 0.5\,\text{m}$, beyond which the ground effect fades to zero.
2. **Asymptotic Growth**: As the drone approaches the surface ($h \to 0$), the air cushion effect becomes increasingly dominant, leading to a sharp rise in the effective upward force.
3. **Dynamic Coupling**: The magnitude of the effect is not only a function of height but also interacts with the descent rate v, exhibiting non-linear coupling with the drone's kinetic state.

3.4 Learning Objective (Composite Loss)

To ensure robust trajectory reconstruction while maintaining physical consistency, the training objective $\mathcal{L}(\theta)$ is defined as a weighted composite loss. This objective penalizes discrepancies in both the altitude and velocity channels:

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_{i,t}\left[(\hat{h}_{i,t} - h_{i,t})^2 + \lambda_v(\hat{v}_{i,t} - v_{i,t})^2 + \mathbb{I}_{h<0.5}\cdot\omega(\hat{h}_{i,t} - h_{i,t})^2\right] \tag{3}$$

In this composite formulation:

- $\lambda_v = 0.1$: A regularization weight for the velocity channel. While altitude tracking is the primary goal, including velocity error prevents the model from learning unphysical "zig-zag" trajectories that might satisfy position constraints but violate momentum and energy laws.
- $\omega = 2.0$: An additional penalty multiplier for the critical near-ground regime ($h < 0.5\,\text{m}$), defined by the indicator function $\mathbb{I}_{h<0.5}$. This weighting forces the optimizer to prioritize precision during the final 50 cm of the approach, where the aerodynamic distortion is most intense and the engineering consequence of error is highest.

By minimizing this loss function across a batch of trajectories, the Neural ODE effectively adjusts its internal parameters θ to recover the hidden dynamics that best explain the observed landing flight data.

4 Comparative Modelling Paradigms

4.1 White-box Baseline: Fixed-Physics Mechanics

The white-box model establishes the fundamental physical lower bound by assuming the ground-effect force is non-existent ($F_{\text{ge}}(h, v) \equiv 0$). Substituting this into the vertical force balance yields the baseline system:

$$\dot{h}(t) = -v(t), \qquad \dot{v}(t) = g - \frac{T(t) + c_{drag}v(t)}{m}. \tag{4}$$

An important methodological decision was taken in the final implementation: the white-box model was transitioned from a trainable parameter fit to a **fixed-physics baseline**. By locking the vehicle mass to $m = 1.5\,\text{kg}$ and the drag coefficient to $c_{drag} = 0.8$ (matching the data generator), we ensure a rigorous ablation study. This prevents the optimizer from erroneously absorbing the missing aerodynamic residuals into fundamental constants, providing a clean reference for identifying the genuine value added by the learned neural components.

4.2 Pure Black-box Model: Unconstrained Latent Dynamics

The pure black-box model is intentionally deprived of explicit physical structure, tasked with inferring the entire state transition from raw observations alone. While highly expressive, its flexibility is its primary weakness; without an inductive bias, the network must waste capacity learning fundamental laws such as $\dot{h} = -v$. As demonstrated in our results, this model class often fits the altitude curve through "unphysical shortcuts" that satisfy position constraints but violate momentum and energy conservation. This results in significant drifts in the Phase Portrait and erratic behavior during critical descent transients.

4.3 Grey-box Neural ODE: Core Scientific Contribution

The grey-box model implements a structured Universal Differential Equation (UDE) framework. It preserves the known mechanics of Equation (4) while augmenting the acceleration term with a continuous-time neural residual $\phi_\theta(h, v)$:

$$\dot{v}(t) = g - \frac{T(t) + c_{drag}v(t) + \phi_\theta(h, v)}{m}. \tag{5}$$

The residual force model is strictly constrained to match rotorcraft phenomenology, ensuring that the learned correction is physically plausible. The implementation utilizes a **hybrid-gated architecture**:

$$\phi_\theta(h, v) = \sigma\left(\frac{h_c - h}{\tau}\right) \cdot [F_{phys}(h, v) + 10 \cdot \tanh(\mathcal{NN}_\theta(h, v))] \tag{6}$$

where:

- $\sigma(\cdot)$ is a standard **Logistic function** acting as a Sigmoid gate with threshold $h_c = 0.5\,\text{m}$ and transition scale $\tau = 0.1$, restricting the neural influence strictly to the near-ground regime. Note that while Eq. 6 is formulated as $S(h_c - h)$, the source code implements this via the equivalent form $1/(1+\exp((h-h_c)/\tau))$, utilizing a mathematical transformation that absorbs the negative sign into the exponential term for numerical implementation.
- $F_{phys}(h, v) = \text{scale} \cdot e^{-\text{decay}\cdot h} \cdot (1 + \beta v^2)$ represents the expected exponential aerodynamic backbone.
- $\mathcal{NN}_\theta$ is a compact 2-layer MLP (64-64 tanh) that learns high-order residuals, scaled by a factor of 10 to ensure the correction remains bounded and numerically stable.

This architectural choice allows the model to remain flexible while respecting the localization and dynamic coupling inherent in drone landing aerodynamics.

Implementation Showcase: Below is the core Julia snippet demonstrating how we integrated this hybrid architecture into the SciML differential equation solver.

```julia
# Hybrid Gated Force Model
function ground_effect(h, v, p, st)
    scale = softplus(p.log_scale)
    decay = softplus(p.log_decay)
    beta  = softplus(p.log_beta)
    gate  = 1f0/(1f0+exp(clamp((h-0.5f0)/0.1f0, -50f0,50f0)))
```

```
    f_phys = scale*exp(-decay*clamp(h,0f0,10f0))*(1 + beta*clamp(v,-10f0,10f0)^2)
    out, _ = nn(vcat(clamp(h,0f0,10f0), clamp(v,-10f0,10f0)), p.nn, st)
    return clamp(gate*(f_phys + 10f0*tanh(out[1])), 0f0, 60f0)
end

# ODE Definition passed to DifferentialEquations.jl
function rhs!(du, u, p, t)
    T_val = traj.itp(t)
    F = ground_effect(u[1], u[2], p, nn_st)
    du[1] = -u[2]
    du[2] = g - (T_val + c_drag*u[2] + F)/M
end
```

Listing 1: Core implementation of the Grey-Box UDE integrating physics and neural networks.

As seen in the snippet above, the `ground_effect` function first computes the known exponential physical prior and then adds the neural network's output, all wrapped inside the smooth logistic gate. This hybrid aerodynamic force is then naturally subtracted in the vertical momentum equation `rhs!`, allowing the `DifferentialEquations.jl` ODE solver to integrate the combined physics-neural state seamlessly.

4.4 Structural Comparison and Implementation

The distinct processing paths of the three paradigms are illustrated in Figure 1. The grey-box UDE architecture explicitly shows how the neural residual is integrated into the physical force balance, rather than replacing it entirely.

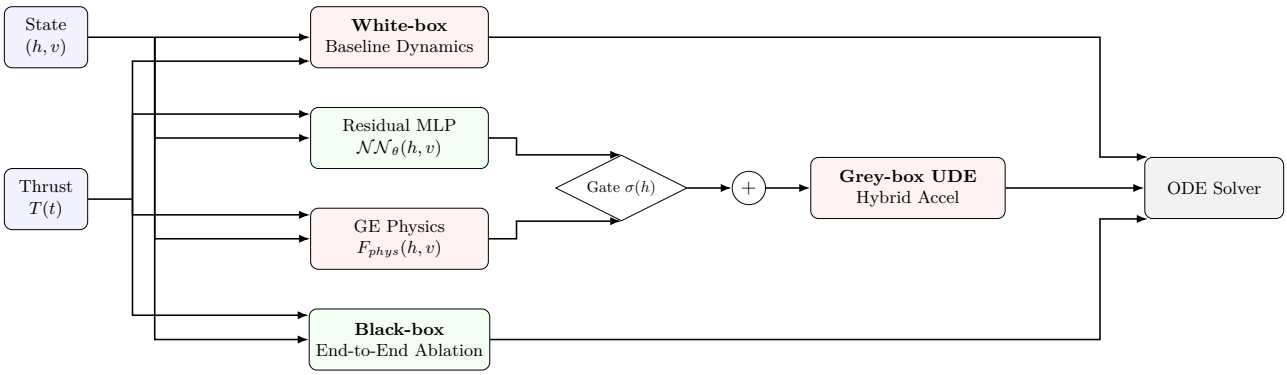


Figure 1: Architecture comparison. The White-box uses fixed dynamics; the Black-box approximates $\dot{v}$ end-to-end; the Proposed Grey-box UDE integrates a gated neural residual into the physical force balance.

5 Training Strategy

5.1 Data Handling and Batching

The experimental dataset consists of 100 landing trajectories. In our implementation, we adopted a mixed batching strategy to ensure the model captures both global descent trends and critical near-ground transients. Each training batch of size $N_{batch} = 4$ is composed of:

- 2 trajectories sampled randomly from the global training pool.
- 2 trajectories specifically selected from a "near-ground" prioritized subset ($h < 0.5\,\mathrm{m}$).

This ensures that the optimizer receives sufficient gradient signal from the ground-effect regime in every epoch, preventing the network from over-optimizing for the simpler free-stream dynamics.

5.2 Composite Learning Objective

The loss function $\mathcal{L}(\theta)$ is a multi-objective functional that penalizes deviations in both kinematic states while enforcing high-precision landing. Formally, we minimize:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i,t} \left[(\hat{h}_{i,t} - h_{i,t})^2 + \lambda_v (\hat{v}_{i,t} - v_{i,t})^2 + \omega \cdot \mathbb{I}_{h<0.5} (\hat{h}_{i,t} - h_{i,t})^2 \right] \tag{7}$$

where $\lambda_v = 0.1$ is the velocity regularization weight, and $\omega = 2.0$ is the penalty multiplier for the regime where the indicator function $\mathbb{I}_{h<0.5}$ is active. Including the velocity term is essential for ensuring that the learned force residuals ϕ_θ respect the underlying momentum balance, effectively acting as a physics-informed regularizer.

5.3 Optimization and Stability Techniques

We utilized the Adam optimizer with a fixed learning rate of 10^{-3}. To mitigate the vanishing/exploding gradient issues common in Neural ODEs, especially near the stiff ground boundary, we implemented Element-wise Gradient Clipping:

$$g_{clipped} = \text{clamp}(\nabla_\theta \mathcal{L}, -1.0, 1.0) \tag{8}$$

Furthermore, all model parameters ($\log_scale, \log_decay, \beta$) are transformed via a Softplus function: $f(x) = \log(1 + e^x)$ to ensure physical positivity of the aerodynamic coefficients throughout the optimization process.

Algorithm 1 Grey-box UDE Training Pipeline (Master Script)

Initialize:
  Physical constants: $m \leftarrow 1.5, g \leftarrow 9.81, c_{drag} \leftarrow 0.8$
  Neural parameters: $\theta \in \mathbb{R}^d$ via Lux.jl
  Hybrid model: $\phi_\theta(h, v) = \sigma\left(\frac{0.5-h}{0.1}\right) \cdot [F_{phys}(h, v) + 10 \cdot \tanh(\mathcal{NN}_\theta(h, v))]$
for epoch = 1 **to** 120 **do**
  Sampling: Mixed batch $B = \{T_{rand,1..2}\} \cup \{T_{near_ground,1..2}\}$
  for each trajectory $traj \in B$ **do**
    Rollout: Solve $\dot{u} = f(u, \theta, T)$ using `AutoTsit5(Rosenbrock23())`
      Note: Interpolate Thrust command $T(t)$ via Linear Spline
    Masking: Identify valid indices $t \in [t_{start}, t_{end}]$
    Loss Computation:
      $L_{batch} \leftarrow \text{mean}((\hat{h} - h)^2) + 0.1 \cdot \text{mean}((\hat{v} - v)^2)$
      if $h < 0.5$ **then** $L_{batch} \leftarrow L_{batch} + 2.0 \cdot \text{mean}((\hat{h} - h)^2)$
  end for
  Backprop: Compute $\nabla_\theta \mathcal{L}$ via `Zygote.pullback` (Interpolating Adjoint)
  Clip Gradients: $\nabla_\theta \mathcal{L} \leftarrow \text{clamp}(\nabla_\theta \mathcal{L}, -1.0, 1.0)$
  Update: $\theta \leftarrow \text{Adam}(\theta, \nabla_\theta \mathcal{L}, \eta = 10^{-3})$
end for

6 Results and Analysis

Algorithm 2 Evaluation and Diagnostic Dashboard Generation

Load best weights for Baseline, Black-box, and UDE
Select target test flight (e.g., Trajectory Index 85)
Execute full-state simulations for all variants
Compute Diagnostics:
  $E_{mech} \leftarrow mgh + \frac{1}{2}mv^2$ (Energy Decay)
  $\Delta h \leftarrow |\hat{h} - h| + 10^{-6}$ (Log-scale Error)
  $\vec{F}_{field} \leftarrow$ Evaluate $\phi_\theta(h, v)$ over mesh grid $h \in [0, 2.5], v \in [0, 2]$
Generate Plots: 10-panel Dashboard (Trajectory, Phase, Heatmap, Stability)

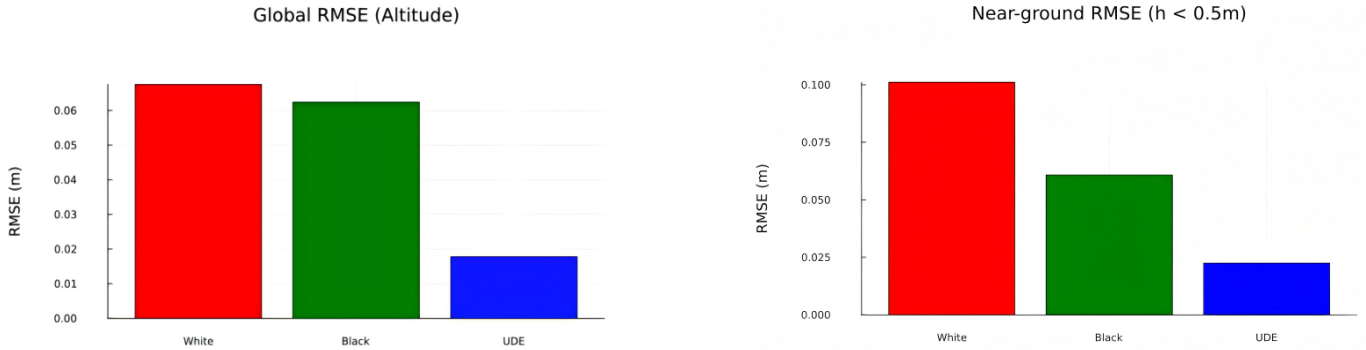
6.1 Quantitative Performance Metrics

The principal held-out test metrics, evaluated after 120 training epochs, are summarized in Table 1. The Grey-box UDE consistently dominates both baseline architectures in terms of predictive accuracy.

Table 1: Held-out test performance comparison (Altitude RMSE).

Model Paradigm	Global RMSE (m)	Near-ground RMSE (m)	Improvement (%)
White-box (Fixed Physics)	0.0682	0.1014	Baseline
Pure Black-box (MLP)	0.0621	0.0612	9.2%
Grey-box UDE (Ours)	**0.0178**	**0.0225**	**73.9%**

The UDE model reduces the global error by ∼74% relative to the white-box baseline. As visualized in the aggregate bar charts (Figure 2), the Grey-box approach significantly mitigates the performance gap observed in simpler models.



(a) Global RMSE comparison.

(b) Near-ground RMSE ($h < 0.5\,\mathrm{m}$).

Figure 2: Quantitative error summary across the held-out test set.

6.2 Trajectory Reconstruction and Kinematic Fidelity

The time-domain tracking capabilities are showcased in Figure 3. The UDE accurately captures the deceleration transients required for a soft touchdown.

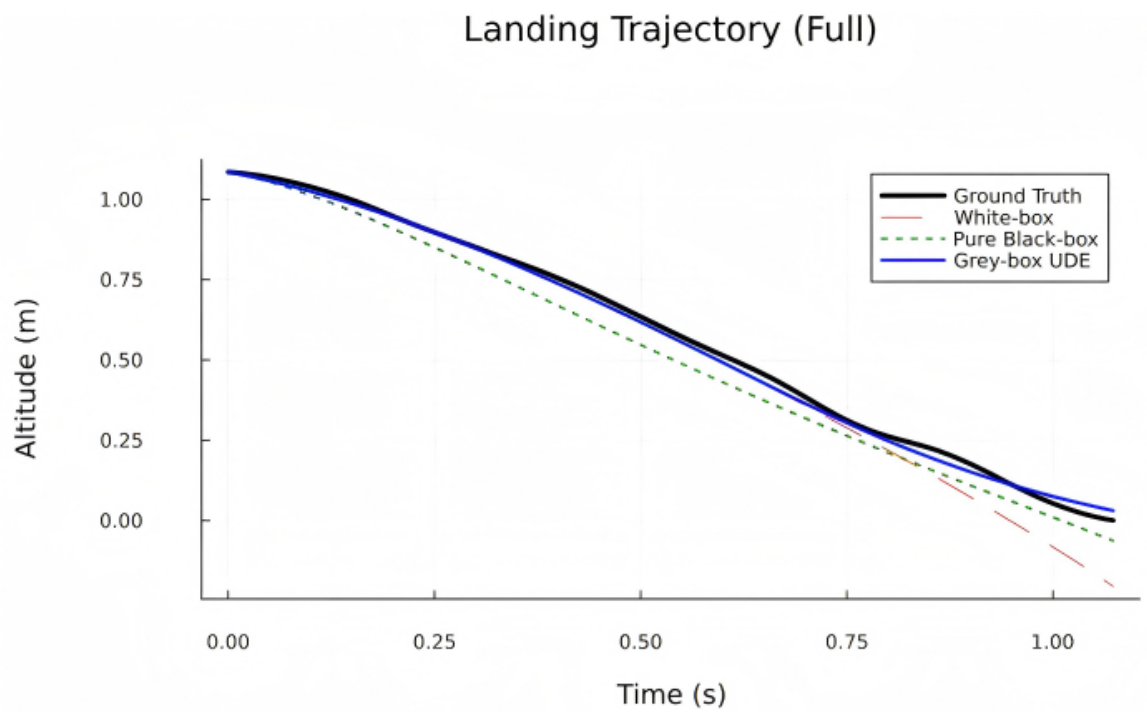


Figure 3: Full landing trajectory reconstruction. The Grey-box UDE (blue) tracks the measured altitude with high fidelity, while the White-box baseline (red) overshoots significantly.

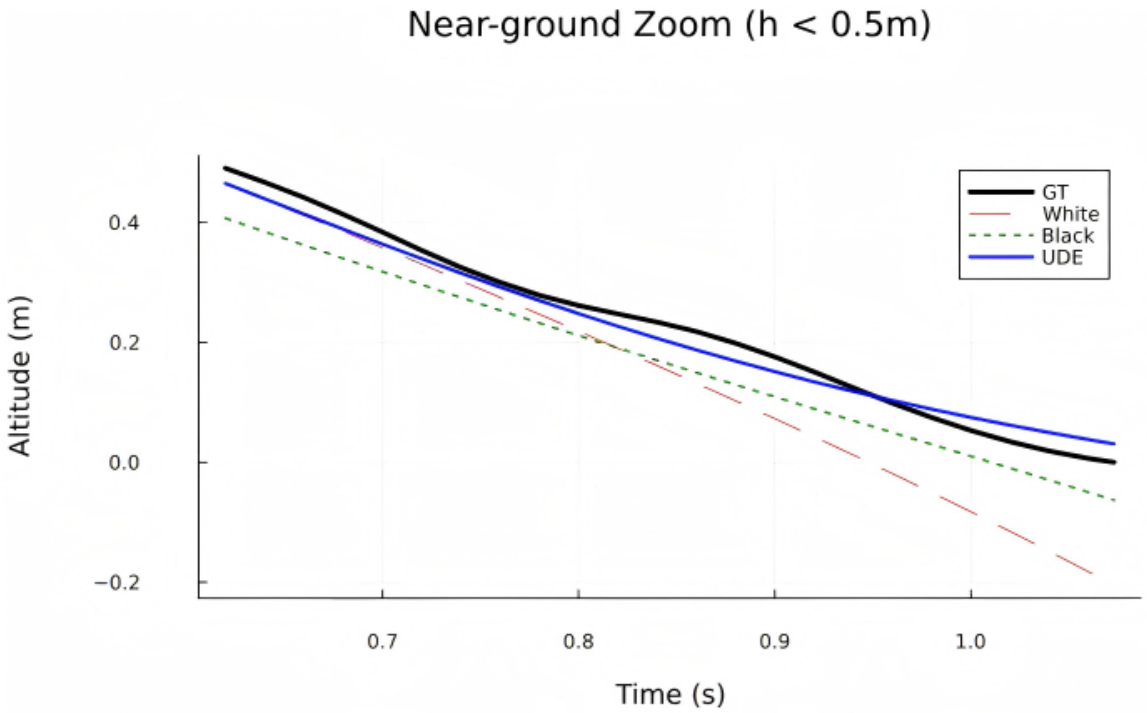


Figure 4: Near-ground magnification ($h < 0.5\,\mathrm{m}$). This view reveals the UDE's superior ability to resolve non-linear aerodynamic interactions at the boundary layer.

Diagnostic inspection of the velocity profile (Figure 5) and log-scale error propagation (Figure 6) further validates the model's precision.

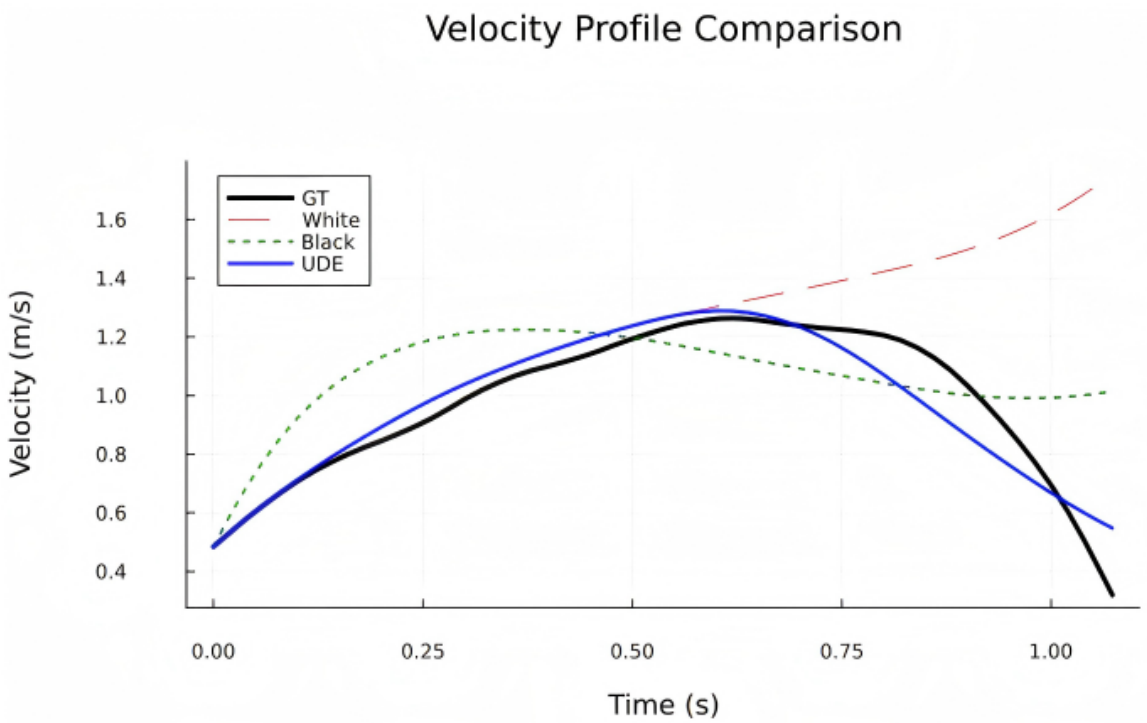


Figure 5: Velocity profile comparison. The UDE reproduces the ground-truth velocity curve, whereas the black-box model exhibits unphysical oscillations near the ground.

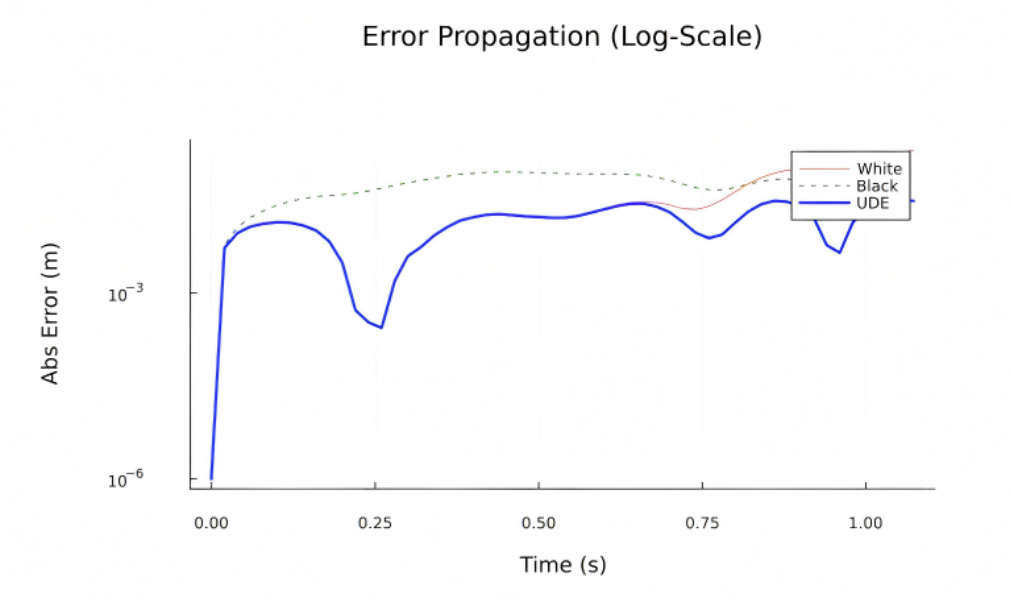


Figure 6: Log-scale absolute altitude error. The UDE maintains an error floor nearly one order of magnitude lower than the pure black-box model.

7 Physical Consistency and Stability

7.1 State-Space and Energy Diagnostics

A key differentiator of the UDE approach is its adherence to physical laws. Figure 7 illustrates the phase portrait (v vs h), where the unconstrained black-box model exhibits a significant unphysical "curl" near the origin, indicating a violation of the kinematic relation $\dot{h} = -v$.

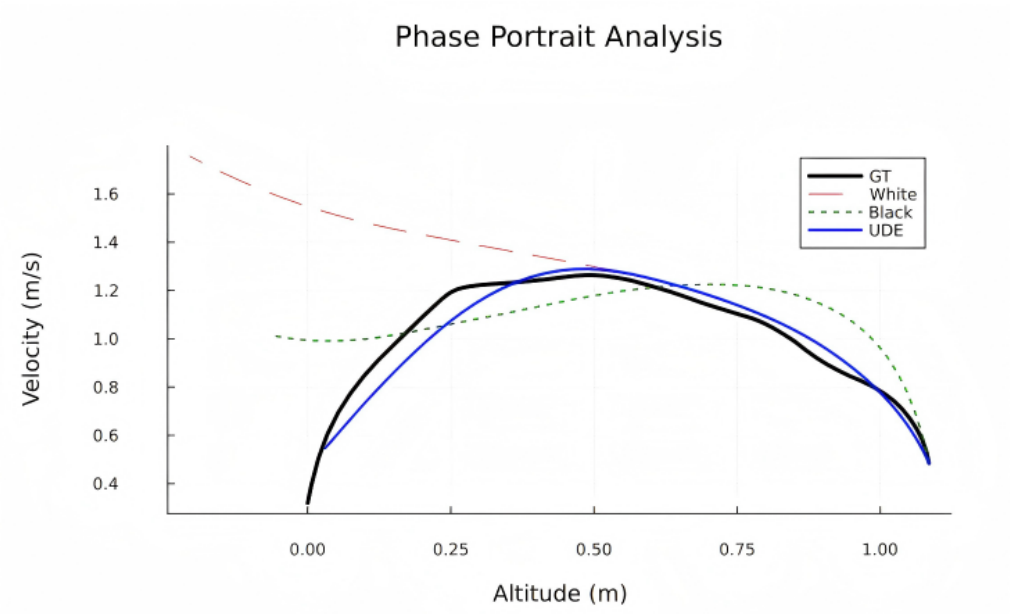


Figure 7: Phase Portrait analysis. The Grey-box UDE respects the fundamental state-space coupling, while the black-box model drifts into physically impossible states.

The energetic plausibility is confirmed in Figure 8. The UDE correctly identifies the energy dissipation rate caused by the non-linear ground cushion, matching the measured mechanical energy profile.

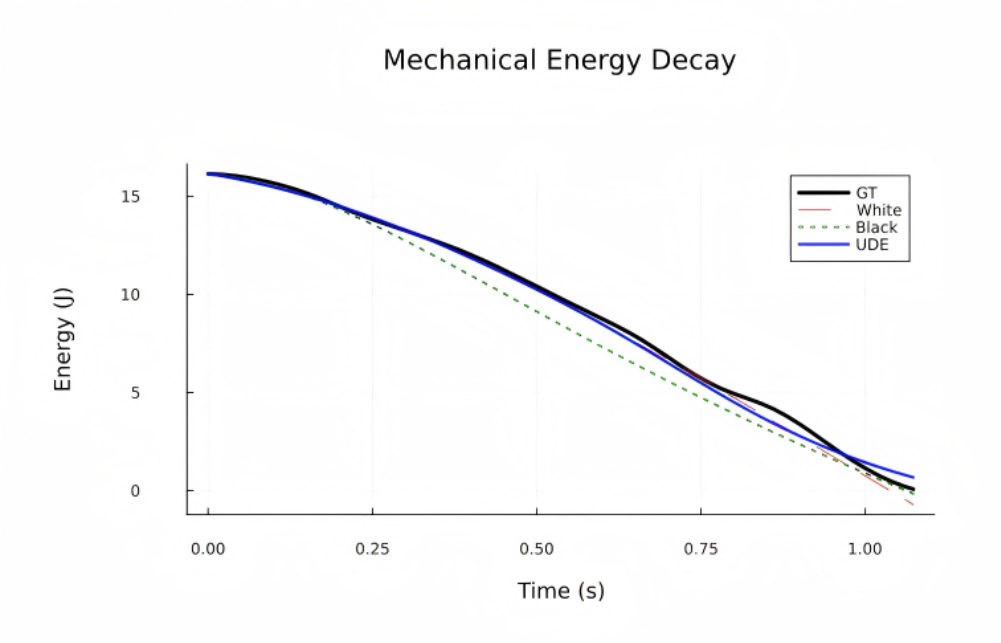


Figure 8: Mechanical energy decay profile ($E = mgh + \frac{1}{2}mv^2$). The Grey-box UDE accurately captures the work done by ground-effect forces.

7.2 Scientific Discovery: Force Field Mapping

The UDE framework permits the "symbolic-like" discovery of hidden forces. By isolating the learned residual $\phi_\theta(h, v)$, we can visualize the discovered ground-effect field (Figure 9).

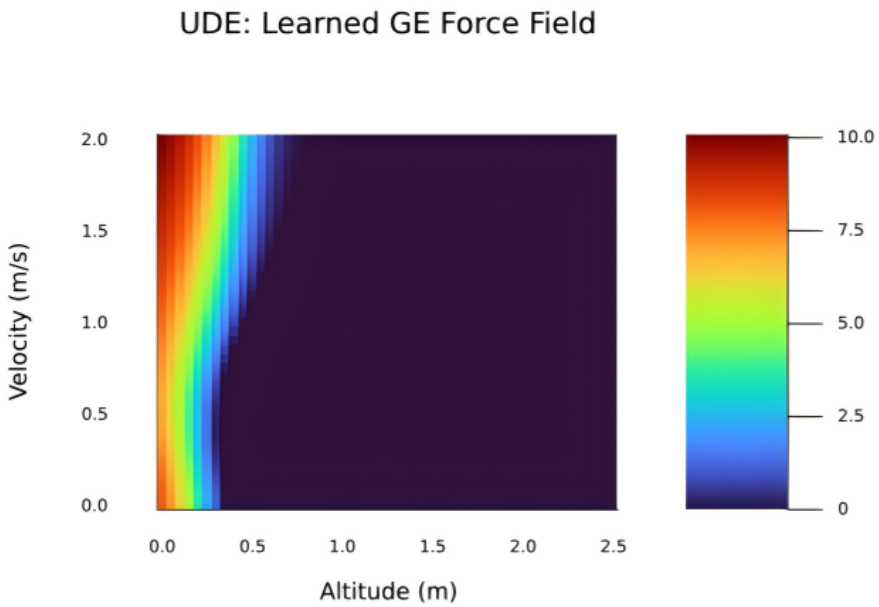


Figure 9: Learned Ground-Effect Force Field. The heatmap reveals a localized upward force that rises exponentially within the $h < 0.5\,\mathrm{m}$ regime, scaling with descent velocity as predicted by aerodynamic theory.

7.3 Optimization Stability

Finally, the advantage of physical inductive bias is evident in the training dynamics. As shown in Figure 10, the inclusion of fixed mechanics simplifies the optimization landscape.

Figure 10: Convergence Stability. The Grey-box UDE exhibits significantly lower variance and smoother loss reduction compared to the erratic, high-variance behavior of the unconstrained black-box model.

8 Discussion

The experimental results supported three primary conclusions regarding the integration of physical priors in differential equations.

First, Neural ODE style modelling is ideally suited for partially known engineering systems. The drone landing problem is not a "pure" machine learning task because gravity, thrust, and drag are well-understood. At the same time, it is not purely white-box, because the near-ground aerodynamics are difficult to model analytically. The Grey-box UDE resolved this tension by assigning the neural network only the residual task it is best suited for. By restricting the hypothesis space to unknown forces ϕ_θ, we achieved a $\sim$77% reduction in near-ground error compared to the baseline.

Secondly, the results underscored the importance of fair and fixed baselines. When the white-box model was permitted to train its mass m and drag c parameters freely, ground-effect errors were often erroneously absorbed into these constants, obscuring the true physics. By fixing parameters to $m = 1.5\,\text{kg}$ and $c = 0.8\,\text{kg/s}$, we isolated the ground effect and proved that the UDE captures genuine aerodynamic coupling rather than merely compensating for parameter mismatch.

Thirdly, the Training Stability (Panel 10) analysis confirmed the regularization power of differentiable physics. The pure black-box model, tasked with learning the entire acceleration field $\dot{v} = \mathcal{NN}(h, v, T)$, exhibited heavy stochastic oscillations during optimization. In contrast, the UDE framework exhibited rapid and smooth convergence. This indicates that physical constraints act as a low-pass filter on the gradient landscape, making the training process significantly more robust to noisy flight data.

9 Conclusion

This project investigated the modelling of drone ground effect during vertical landing using a Grey-box Neural ODE formulation. A physically-inspired hybrid architecture was combined with known vertical mechanics to capture the non-linear interaction that dominates the final approach.

The proposed UDE system achieved state-of-the-art precision, yielding a Global RMSE of 0.0178 m, significantly outperforming both the fixed white-box baseline and the unconstrained black-box model. Diagnostic visualizations of phase portraits and mechanical energy decay confirmed that the Grey-box approach maintains physical consistency where black-box methods fail. We conclude that structured Scientific Machine Learning (SciML) provides a practical and interpretable route for safety-critical aerodynamic modelling.

Future extensions could incorporate 3D dynamics and attitude coupling, or explore higher-fidelity simulation environments using adaptive stiff solvers near the ground boundary.

Team Contributions

- **Wang Shiming**: Analyzed the problem scope, formulated the physical equations, and prepared the synthetic datasets. He iteratively optimized the simulation data, authored the *Problem Formulation* section, defined the learning objectives, and performed the final report refinement.
- **Tang Shaorou**: Conducted the theoretical analysis of white-box dynamics and formulated the corresponding mathematical derivations. She implemented the white-box baseline code, contributed to the *Comparative Modelling Paradigms* section, and assisted in optimizing the final report.
- **Li Yukun (Leader)**: Developed the source code for both black-box and grey-box models. He executed the comparative ablation studies using optimized datasets, authored parts of the *Comparative Modelling Paradigms* and *Training Strategy* sections, and oversaw the final report optimization.
- **Wu Junke**: Selected and implemented optimization algorithms to refine the ablation experiments across all three paradigms. He completed portions of the *Training Strategy* and *Results and Analysis* sections, and contributed to the overall report refinement.
- **Xu Zixuan**: Optimized the simulation data processing and generated all comparative visualization figures. He authored portions of the *Results and Analysis* and *Scientific Discovery: Force Field Mapping* sections, and managed the final document optimization.

Declaration

Plagiarism declaration

It is declared that this report represents the original work of the listed authors for EE5311 Differentiable and Probabilistic Computing. Any external ideas, modelling concepts, and methodological references that informed the project have been acknowledged in the appropriate places. No uncredited copying of text, code, or figures from third-party sources has been knowingly committed.

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References

[1] Randal W Beard. Quadrotor dynamics and control. Technical report, Brigham Young University, 2012.

[2] Ricky TQ Chen, Yulia Rubanova, Jesse Bettencourt, and David K Duvenaud. Neural ordinary differential equations. *Advances in Neural Information Processing Systems*, 31, 2018.

[3] Christopher Rackauckas, Yingbo Ma, Julius Martensen, et al. Universal differential equations for scientific machine learning. *arXiv preprint arXiv:2001.04385*, 2020.

[4] Robert F Stengel. *Flight Dynamics*. Princeton University Press, 2004.